RF Scanner Based on Rydberg Sensor

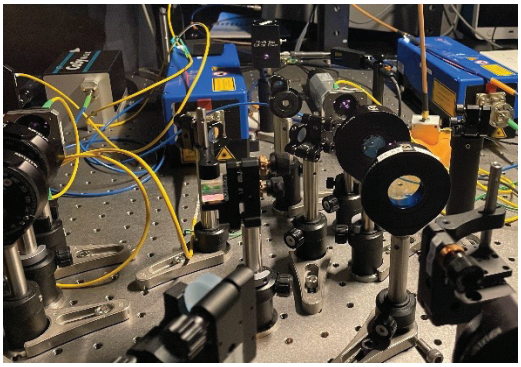
Product Name: RF Scanner based on Rydberg Sensor
Category: Tabletop Prototype of RF Field Measurement Device
Technology: Quantum-based E-Field Sensing using Rydberg Atoms

Product Description

The RF Scanner based on Rydberg Sensor is a cutting-edge RF E-field measurement device based on room temperature Rubidium atoms designed to provide precise and SI-traceable measurements of electromagnetic fields in the RF and microwave range. Leveraging the quantum properties of Rubidium atoms excited to Rydberg state, our proposed RF Scanner offers exceptional sensitivity and broad tunability, capable of accurately measuring electric fields over different frequency bands without the need for traditional RF calibration standards. Compared to the other existing methods, our proposed system will only use two near-infrared lasers to exploit the quantum properties of Rydberg atoms for high precision modern RF measurements competing well with the existing sensors. With a quoted sensitivity of ~μV/cm/$\sqrt{Hz}$ it has the capability of an improved sensitivity of ~ nV/cm/$\sqrt{Hz}$ with some advanced tweaks in the system. The same sensor can as well work as an atomic RF receiver to receive Radio frequency communication signals.

Features

- Broadband tunability
- High sensitivity
- Calibration-free Operation
- Real-time measurement
- Calibrate RF antennas
- Electromagnetic Compatibility (EMC) testing



Representative Optical Setup

Technical Specifications

Parameter	Specification
Frequency Range	Sub-GHz UHF band (600MHz to 1 GHz) Ku band (15 to 18 GHz) K-band (18 GHz to 26.5 GHz) Ka-band (26 GHz to 40 GHz)
E-field Sensitivity	~μV/cm/$\sqrt{Hz}$
Minimum E-field	~mV/cm
Atomic Response	Resonant
Detection Method	Standard EIT-ATS/EIA-ATS technique
Atom Species	Rubidium
Rydberg Level Tuning	Changing 1260 nm laser wavelength
Calibration Period	None (Self-Calibrated)
Laser Stabilization	Two photon Spectroscopy
Power Supply	Dc power supply or AC Input
Output Port	BNC